

Qingdao Workshop on Coding Theory and Related Topics

Program and Abstracts

Date: June 22–26, 2026

Location: Lecture Hall (N510), Huagang North Building
Shandong University Qingdao Campus, Qingdao, China

会议地点：山东大学（青岛校区）华岗苑北5楼报告厅N510

Organizers

Weijun Fang, Sihuang Hu, Guodong Li
Chong Shangguan, Yiwei Zhang

Hosted by

Research Center for Mathematics and Interdisciplinary Sciences
School of Cyber Science and Technology
Shandong University

Qingdao Workshop on Coding Theory and Related Topics

Schedule

June 22–26, 2026

Monday, June 22

Time	Speaker	Title
Morning Host: Sihuang Hu		
09:00–09:20	Chong Shangguan	Opening
09:20–10:10	Alexander Barg	Classical and Quantum Reed–Muller Codes and Their Extensions
10:10–10:40	Tea Break	
10:40–11:30	Pavel Panteleev	Expansion in Tensor Products of Codes over Large Fields
Afternoon Host: Guodong Li		
14:30–15:20	Zimu Li	Fault-Tolerant Logical Operations on Quantum LDPC Codes
15:20–15:50	Tea Break	
15:50–16:40	Hedongliang Liu	Affine Subcode Ensemble Decoding for Degeneracy-Aware Quantum Error Correction
16:40–17:00	Open Problems	

Tuesday, June 23

Time	Speaker	Title
Morning Host: Yiwei Zhang		
09:20–10:10	Tuvi Etzion	Binary and Non-Binary Self-Dual Codes and Maximum Period Single-Track Gray Codes
10:10–10:40	Tea Break	
10:40–11:30	Zitan Chen	Deletion-Correcting Codes for an Adversarial Nanopore Channel
Afternoon Host: Yiwei Zhang		
14:30–15:20	Xue Chen	Algorithms for Sparse LPN and LSPN Against Low-Noise
15:20–15:50	Tea Break	
15:50–16:40	Hsin-Po Wang	Not-So-Perfect Hashing for Massive Random Access
16:50–17:40	Wei-Hsuan Yu	Semidefinite Programming Bounds for Designs on Spheres and Hamming Spaces

Wednesday, June 24

Time	Speaker	Title
Morning Host: Chong Shangguan		
09:20–10:10	Zeyu Guo	Explicit Rank Extractors and Subspace Designs via Function Fields, with Applications to Strong Blocking Sets
10:10–10:40	Tea Break	
10:40–11:30	Nicolas Resch	Discrepancy Problems for Linear Spaces or Decoding Problems above Capacity

Thursday, June 25

Time	Speaker	Title
Morning Host: Weijun Fang		
09:20–10:10	Xiangliang Kong	Some Recent Progress on Convertible Codes for Distributed Storage
10:10–10:40	Tea Break	
10:40–11:30	Kuan Cheng	Weighted Pseudorandom Generators for Read-Once Branching Programs via Weighted Pseudorandom Reductions
Afternoon Host: Weijun Fang		
14:30–15:20	Nicolas Resch	Fast Codes for Fast Cryptography
15:20–15:50	Tea Break	
15:50–16:40	Chen Yuan	Faster Pseudorandom Correlation Generators from Quasi-Abelian Codes
16:40–17:00	Open Problems	

Friday, June 26

Time	Speaker	Title
Morning Host: Chong Shangguan		
09:20–10:10	Jiaxi Nie	Evasive Sets, Twisted Varieties, and Container-Clique Trees
10:10–10:40	Tea Break	
10:40–11:30	Yiwei Zhang	A Disguise-and-Squeeze Approach for MDS-TPIR (Private Information Retrieval)

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Classical and Quantum Reed–Muller Codes and Their Extensions

Alexander Barg

University of Maryland

abarg@umd.edu

Abstract: Explicit description of logical operators of well-known families of quantum codes has been addressed in numerous recent works. We present a geometric description of a set of transversal logical operators of quantum Reed–Muller codes. In the second part we extend the construction of the classical RM family to binary codes defined on an arbitrary finite Coxeter complex and establish basic properties of the new code class, including the parameters of CSS quantum codes obtained from Coxeter codes.

Note: This is based on joint work with Nolan Coble, Dominik Hangleiter, and Christopher Kang.

Keywords: Quantum Reed–Muller codes; Transversal logical operators; Coxeter complex; CSS quantum codes.

Expansion in Tensor Products of Codes over Large Fields

Pavel Panteleev

Moscow State University , Russia

panteleev@intsys.msu.ru

Abstract: Tanner codes give a powerful way to construct LDPC codes from sparse graphs by imposing local code constraints at the vertices. When the local code has large distance and the underlying graph is a good expander, the resulting code has linear rate, linear distance, and efficient decoding. Recent progress on quantum LDPC codes and locally testable codes has led to higher-dimensional analogues of this picture, where graphs are replaced by cubical complexes and the local constraints are built from tensor products of codes.

A key difficulty in the higher-dimensional setting is that large minimum distance of the local codes is no longer the right condition. It must be replaced by a stronger parameter, called expansion, which can be viewed as a higher-dimensional analogue of minimum distance. In this talk, I will present a new result showing that, over a sufficiently large finite field, almost all tensor products of codes have expansion linear in their length. This provides the local building blocks needed for high-dimensional Tanner-type constructions in arbitrary dimension, with applications to locally testable codes, complexity theory, and fault-tolerant quantum computation.

Fault-Tolerant Logical Operations on Quantum LDPC Codes

Zimu Li

Yau Mathematical Sciences Center, Tsinghua University

zimu0708@gmail.com

Abstract: Logical quantum operations, with non-Clifford ones being the key bottleneck, are a central problem in fault-tolerant quantum computing parallel to the quest for qLDPC codes with desirable parameters. While recent breakthroughs have achieved good qLDPC codes and almost-good qLTCs, logical gates on such quantum codes remain severely restricted and little understood.

In this talk, we introduce our recent work on the formation of fault-tolerant logical gates. We first show fundamental no-go theorems for transversal non-Clifford gates on product-type codes. Then, we present a unified framework for constructing and certifying constant-depth non-Clifford gates on almost-good qLDPC codes and qLTCs. Our approach analyzes (co)homological invariants of quantum codes, from hypergraph product (HGP) codes to sheaf codes that encompass known good qLDPC codes. Specifically, we interpret sheaf codes as covering spaces of HGP codes via graph lifts. This yields asymptotic code families generated from constant-size seeds while preserving the invariants, so logical gates can be verified on small instances and lifted to large codes. Using this framework, we introduce cupcap gates, a new family of homological invariant forms inducing constant-depth nontrivial logical multi-controlled-Z gates. We also extend canonical logical representatives to sheaf codes, resolving a long-standing challenge in explicitly characterizing sheaf codewords. We prove the existence of $\tilde{\Theta}(N)$ independent cup products under standard number-theoretic conjectures, opening the door to linearly many parallel constant-depth nontrivial multi-controlled-Z gates. These results uncover and highlight the deep interaction between quantum coding theory and mathematics. If time permits, we may also introduce our recent relevant work on non-Abelian codes.

Note: This presentation is based on the following arXiv references: 2507.16797, 2512.21922, 2603.25831, and 2604.01874.

Keywords: Quantum LDPC codes; Fault-tolerant logical gates; Hypergraph product codes; Sheaf codes; Homological invariants.

Affine Subcode Ensemble Decoding for Degeneracy-Aware Quantum Error Correction

Hedongliang Liu

Karlsruhe Institute of Technology

hedongliang.liu@kit.edu

Abstract: Quantum low-density parity-check codes are promising candidates for fault-tolerant quantum computing, but degeneracy is known to impair the convergence of belief-propagation (BP) decoding. In this work, we show that appending linearly independent rows to a check matrix of a stabilizer code can reduce the search space for valid degenerate solutions. We extend the recently proposed affine subcode ensemble decoding technique from the classical to the quantum setting.

Binary and Non-Binary Self-Dual Codes and Maximum Period Single-Track Gray Codes

Tuvi Etzion

Technion – Israel Institute of Technology

etzion@cs.technion.ac.il

Abstract: Binary self-dual sequences have been considered and analyzed throughout the years, and they have been used for various applications. Motivated by a construction for single-track Gray codes, we examine the structure and recursive constructions for binary and non-binary self-dual sequences. The feedback shift registers that generate such sequences are discussed. The connections between these sequences and maximum period single-track codes are discussed. Maximum period non-binary single-track Gray codes of length p^t and period p^{p^t} are constructed. These are the first infinite families of maximum period codes presented in the literature.

Deletion-Correcting Codes for an Adversarial Nanopore Channel

Zitan Chen

The Chinese University of Hong Kong

chenztan@cuhk.edu.cn

Abstract: We study deletion-correcting codes for an adversarial nanopore channel in which at most t deletions may occur. We show that the optimal redundancy of q -ary codes of length n for this channel lies between $t \log_q n + \Omega(1)$ and $2t \log_q n - \log_q \log_2 n + O(1)$. Furthermore, we give an explicit construction of q -ary codes whose redundancy matches the existential upper bound to first order. Finally, we present a second explicit construction that improves on the existential upper bound when $t \leq 2$.

Keywords: Deletion-correcting codes; Nanopore channels; Adversarial errors.

Algorithms for Sparse LPN and LSPN Against Low-noise

Xue Chen

University of Science and Technology of China

xuechen1989@ustc.edu.cn

Abstract: We consider sparse variants of the classical Learning Parities with random Noise (LPN) problem. Our main contribution is a new algorithmic framework that provides learning algorithms against low-noise for both Learning Sparse Parities (LSPN) and sparse LPN problems. Different from previous approaches, this framework has a simple structure without fast matrix multiplication or tensor methods.

Our framework provides a LSPN algorithm that runs in time $O(\eta \cdot n/k)^k$ for any noise rate η , improving the state-of-the-art whenever $\eta \in (k/n, \sqrt{k/n})$. For the sparse LPN problem, we provide a learning algorithm that improves previous approaches in a wide range of parameters, notably for cases like $k = 3$ and $m = n^{1.4}$.

Keywords: Learning parity with noise.

Not-So-Perfect Hashing for Massive Random Access

Hsin-Po Wang

National Taiwan University

hsinpo@ntu.edu.tw

Abstract: This talk discusses how perfect hashing and its not-so-perfect variants can help save communication cost in massive random access (MRA).

For downlink MRA, we borrow the idea of perfect hashing, but allow collisions among users assigned the same message. Compared with the 2025 TIT work of Song, Attiah, and Yu, our scheme increases the probability of success, reduces the expected number of trials, and reduces communication cost. We then invoke the principle of maximum entropy (POME) to conjecture that our upper bound is tight.

For uplink MRA, we propose (α, β) -perfect hashing, where α is the fraction of slots occupied by exactly one user, and β is the fraction of users that are alone in their assigned slot. This generalizes Song and Telatar's α -perfect hashing presented at ISIT 2026. We again use POME to find a competitively tight upper bound on the communication cost, which outperforms and generalizes the bound by Song and Telatar.

Semidefinite Programming Bounds for Designs on Spheres and Hamming Spaces

Wei-Hsuan Yu

National Central University

whyu@ncu.edu.tw

Abstract: We use semidefinite programming methods for spherical codes and energy minimization to improve lower bounds for designs on spheres and Hamming spaces in low-dimensional cases. In particular, we reproduce the optimality of current Hamming designs (or orthogonal arrays, equivalently) and the harmonic index 4-designs in S^2 .

Explicit Rank Extractors and Subspace Designs via Function Fields, with Applications to Strong Blocking Sets

Zeyu Guo

The Ohio State University

zguotcs@gmail.com

Abstract: We give new explicit constructions of several fundamental objects in linear-algebraic pseudorandomness and combinatorics, including lossless rank extractors, weak subspace designs, and strong s -blocking sets over finite fields.

Our focus is on the small-field regime, where the field size depends only on a secondary parameter (such as the rank or codimension) and is independent of the ambient dimension. This regime is central to several applications, yet remains poorly understood from the perspective of explicit constructions.

In this setting, we obtain the first explicit constructions of lossless rank extractors and weak subspace designs for $r \ll k$, where r denotes the rank (or codimension), over finite fields $\mathbb{F}_q$ with $q \geq \text{poly}(r)$ and q non-prime, with near-optimal parameters. For other finite fields, including prime fields and small fields, we obtain weaker but still improved bounds.

As a consequence, we construct explicit strong s -blocking sets in $\text{PG}(k-1, q)$ of size $O(s(k-s)q^s)$ for all sufficiently large non-prime fields $q \geq \text{poly}(s)$, matching the best known non-explicit bounds up to constant factors. This significantly improves the previous best bound $2^{O(s^2 \log s)} q^s k$ of Bishnoi and Tomon (Combinatorica, 2026), which requires $q \geq 2^{\Omega(s)}$.

Our approach is primarily algebraic, combining techniques from function fields and polynomial identity testing. In addition, we develop a complementary Fourier-analytic framework based on ε -biased sets, which yields improved explicit constructions of strong s -blocking sets over small fields.

Note: This is based on joint work with Roshan Raj, Chong Shangguan, and Zihan Zhang.

Keywords: Rank extractors; Subspace designs; Function fields.

Discrepancy Problems for Linear Spaces or Decoding Problems above Capacity

Nicolas Resch

University of Amsterdam

n.a.resch@uva.nl

Abstract: The S -deviation of a linear code C (a k -dimensional subspace of $\mathbb{F}_q^n$) measures how much the intersection $|C \cap B|$ deviates from its expectation over a random choice of C , maximized over all B in a family of sets S . This framework offers a unified way to view fundamental problems in coding theory: when S is the family of Hamming balls of a given radius, we are studying list-decoding. When S is the family of combinatorial rectangles of a given side length, we are studying list-recovery.

While these properties are typically analyzed in the below-capacity regime, we focus on the above-capacity setting, where the expected number of code-words in a ball or rectangle grows exponentially in n . This regime is naturally motivated by applications in secret sharing, hashing, and the study of pseudorandomness, yet it remains significantly less understood than the below-capacity case.

In this talk, I will discuss ongoing research into these problems. In particular, we prove that random linear codes achieve optimal list-decodability and list-recoverability discrepancy in many above-capacity regimes.

Note: Joint work with Dean Doron, Tal Leonov, Henrique Campos Navas, Nicolas Resch and João Ribeiro.

Some Recent Progress on Convertible Codes for Distributed Storage

Xiangliang Kong

Bilkent University

rongxikong@gmail.com

Abstract: In large-scale distributed storage systems, erasure coding is employed to ensure reliability against disk failures. Recent work by Kadekodi *et al.* demonstrates that adapting code parameters to varying disk failure rates can lead to significant storage savings without compromising reliability. Such adaptations, known as *code conversions*, motivate the design of *convertible codes*, which enable efficient transformations between codes with different parameters.

In this talk, I will first introduce the formal framework of code conversion and the corresponding notion of convertible codes proposed by Rashmi *et al.* I will then summarize the known bounds on the access and bandwidth costs for the main classes of convertible codes, along with existing constructions achieving optimal access and bandwidth costs. Next, I will present several new constructions of both scalar and vector MDS convertible codes with optimal access and bandwidth properties. Finally, I will conclude the talk with several possible directions for future research.

Weighted Pseudorandom Generators for Read-Once Branching Programs via Weighted Pseudorandom Reductions

Kuan Cheng

Peking University

ckkcdh@pku.edu.cn

Abstract: I will first describe a recent research line on derandomizing Logspace Randomized Computations, especially about weighted pseudorandom generators (WPRGs), derandomizing algorithms, and their variants in different settings. Then I will introduce our recent results on this line, including improved WPRGs for standard read-once branching programs (ROBPs) and permutation ROBPs, and new derandomizations for regular ROBPs.

Note: The talk is based on a joint work with Ruiyang Wu on SODA 2026.

Keywords: Pseudorandomness; Derandomization; Space Bounded Computation.

Fast Codes for Fast Cryptography

Nicolas Resch

University of Amsterdam

n.a.resch@uva.nl

Abstract: Motivated by the goal of designing highly efficient cryptographic schemes (particularly in the area of secure multiparty computation), we consider the task of designing linear error-correcting codes (or, more precisely, distributions over codes) that are both fast — that is, encoding can be completed in linear time — and additionally achieve the Gilbert-Varshamov (GV) bound with high probability. Furthermore, these cryptographic tasks require us to guarantee that the dual code achieves the same properties of fast encodability and distance at the GV bound.

In this talk, I will introduce two such constructions. The first is inspired by repeat-multiple-accumulate (RMA) codes [1], a class of binary turbo codes which admit very fast encoding. Subsequently I will describe another construction based on codes of Druk & Ishai [2], that allows for generalizing the arbitrary alphabets and rates (the first construction is limited to rate $1/2$), among other benefits, albeit at the cost of poorer concrete efficiency.

Note: Based on a joint works with Martijn Brehm and Yuval Ishai. The first paper is already available on arXiv: 2509.07639, the second one still needs to be uploaded (I hope to take care of that soon).

Faster Pseudorandom Correlation Generators from Quasi-Abelian Codes

Chen Yuan

Shanghai Jiao Tong University

chen_yuan@sjtu.edu.cn

Abstract: The past few years have witnessed the growing importance of pseudorandom correlation generators (PCGs) that allow for generating correlated randomness with sublinear communication. To date, efficient PCGs for (vector) oblivious linear evaluation (OLE) over arbitrary fields have been constructed under syndrome decoding problems, and moreover implemented with practical efficiency for fields of small size, mostly binary ones. However, there remains a huge efficiency gap for PCGs over large prime fields, in particular, the popular Mersenne prime field. The primary computational bottleneck essentially stems from the absence of a linear code that simultaneously has a very fast encoding (more precisely, syndrome computing) algorithm and good minimum distance.

In pursuit of practical efficiency, we give extremely fast encoding/syndrome computing algorithms for quasi-abelian (QA) codes, a well-known family of codes that approach Gilbert-Varshamov (GV) bound. Our algorithms make use of Walsh-Hadamard transform and work for all QA codes over group ring $\mathbb{F}_p[\mathbb{Z}_2^n]$. We also derive a concrete large distance bound for our QA codes over large prime field.

We then apply our techniques in PCG scenarios. For vector-OLE over a 64-bit prime field, we fully implement the construction based on the libOTe library, showing a generation speed $> 5,000,000$ vector-OLEs per second, which outperforms EA/EC based constructions by around 4 times. As for OLE over a 64-bit prime field, we compare Walsh-Hadamard transform with Fast Fourier Transform involved in the Ring-LPN based PCG (Boyle et al. CRYPTO'20). Our experiments show one magnitude of acceleration, eliminating the domination of FFT evaluations in OLE generation.

Evasive Sets, Twisted Varieties, and Container-Clique Trees

Jiaxi Nie

Georgia Institute of Technology

jnie47@gatech.edu

Abstract: In the affine space $\mathbb{F}_q^n$ over the finite field of order q , a point set S is said to be (d, k, r) -evasive if the intersection between S and any variety, of dimension k and degree at most d , has cardinality less than r . As q tends to infinity, the size of a (d, k, r) -evasive set in $\mathbb{F}_q^n$ is at most $O(q^{n-k})$ by a simple averaging argument. We exhibit the existence of such evasive sets of sizes at least $\Omega(q^{n-k})$ for much smaller values of r than previously known constructions, and establish an enumerative upper bound $2^{O(q^{n-k})}$ for the total number of such evasive sets. The existence result is based on our study of twisted varieties. In the projective space $\mathbb{P}^n$ over an algebraically closed field, a variety V is said to be d -twisted if the intersection between V and any variety, of dimension $n - \dim(V)$ and degree at most d , has dimension zero. We prove an upper bound on the smallest possible degree of twisted varieties which is best possible in a mild sense. The enumeration result includes a new technique for the container method which we believe is of independent interest. To illustrate the potential of this technique, we give a simpler proof of a result by Chen–Liu–Nie–Zeng that characterizes the maximum size of a collinear-triple-free subset in a random sampling of $\mathbb{F}_q^2$ up to polylogarithmic factors.

Note: This is joint work with Jack Lim and Ji Zeng.

A Disguise-and-Squeeze Approach for MDS-TPIR (Private Information Retrieval)

Yiwei Zhang

Shandong University

ywzhang@sdu.edu.cn

Abstract: In the MDS-TPIR model, M files are stored across N servers via the same (N, K) -MDS code. A user wants to retrieve a single file without disclosing the index of the file to any set of up to T colluding servers. The rate of a PIR scheme is defined as the ratio of the size of the desired file to the total download. Freij-Hollanti et al. raised a conjecture about the MDS-TPIR capacity (maximum achievable rate), which was later disproved by Sun and Jafar using a counterexample for $(M, N, T, K) = (2, 4, 2, 2)$. Generalizing this counterexample, we propose a new MDS-TPIR scheme via a disguise-and-squeeze approach, which beats the state-of-the-art result for $(M, N, T, K) = (2, N, 2, K)$, $N \geq K + 2$. Moreover, our approach can be adapted to other models such as the multi-file retrieval and PIR with arbitrary collusion patterns. Joint work with Rui Sun, Ran Tao, and Jingke Xu.